

```
print("Computer Programming 2026\n")  
print("Department of Geography\n")  
print("26 February 2026\n")  
  
print("Week 1")
```

Computer Programming

- Why a course on computer programming?
- Who is teaching us about computer programming?
- Info: Examination and Office Hours



Computer Programming

- **Why a course on computer programming?**
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A close-up illustration of Super Saiyan Goku from the Dragon Ball series. He has spiky blonde hair and a determined, shouting expression with his mouth wide open. His right fist is clenched and raised near his chest. The background is a bright, hazy green and yellow with sparkling light effects.

**TO BECOME A SUPER
PROGRAMMER**

Objectives

- Information and how to process information
- Communication between humans and calculators



The ambiguous concept of information

- Meaning 1: [information = data + instructions]

set of data to analyze, and instructions for processing such data

(for example, in the operation $1+2$, the numbers '1' and '2' and the operator '+' are information entities)

- Meaning 2: [information = data with meaning]

what we obtain from the data, after a meaning is given to such data

(for example, the numbers '1' and '2' are simple data; they become information entities only when we decide they refer to apples, airports, and so on...)

Information processing

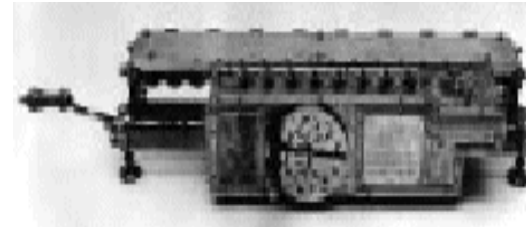
- Every human processes a big amount of information every day without using sophisticated tools
- Two situations make this activity more difficult:
 - when the problem exceeds a certain degree of complexity
 - when it is necessary to process information with substantial speed
- Over the history, humans created many tools to help themselves with information processing

The first calculators...

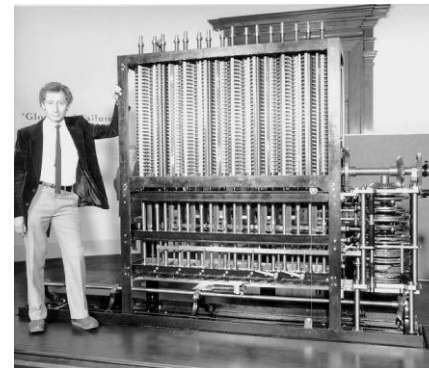
- **Blaise Pascal (1623-1662)**
mechanical device (gears controlled by a crank) for executing addition and subtraction



- **Gottfried Wilhelm von Leibniz (1646-1716)**
introduces also multiplication and division
(four-function calculator)



- **Charles Babbage (1792-1871)**
designs and creates a “**difference engine**”
 - Calculates tables of numbers useful for navigation
 - **only one** algorithm: **finite-difference polynomial**
 - output: **holes** on a copper plate (punch cards)



The calculator: interpreter and executor

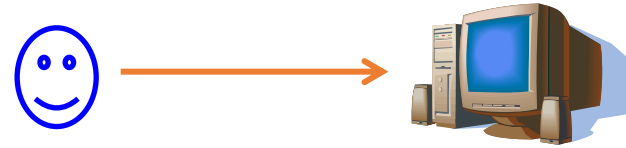
When receiving a request from the user, the calculator performs two activities:

- It checks if it is able to interpret the command (identify the command as correct and related to an action that it is able to execute)
- If such check gives a positive result, it executes the action described by the command, and, when requested, displays the outcome to the user.

Instead of sending one command at a time to the calculator, the user can choose to create a sequence of commands (= program) and send it to the calculator, which is autonomously able to interpret and execute the program, one instruction at a time.

Communication with the calculator

The user-programmer and the calculator must speak the same language

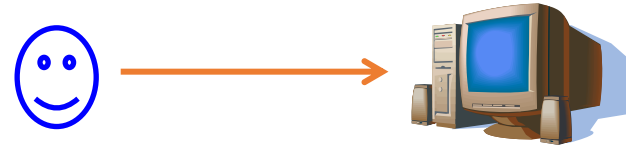


An option may be to “teach the calculator” to understand (= interpret and execute commands expressed in) a natural language such as English or Italian.

- **Advantages:**
 - Semantically-rich languages, able to express commands for sure
 - Languages already known by the user
- **Disadvantages:**
 - Semantically-rich languages, so at risk of ambiguity
 - Complex languages to teach / learn

Communication through formal languages

The user-programmer and the calculator must speak the same language



An alternative option is to create a “programming” language, designated to the communication with the calculator.

- **Advantages:**
 - Specifically-designed language, so very efficient
 - Non-ambiguous language
- **Disadvantages:**
 - Formalized language, so structurally different from the natural languages
 - Language unknown to the user

Week	Date	Topic
Week 1	26/2	Course Introduction
Week 2	5/3	Algorithms
Week 3	12/3	Variables & Conditional Execution
Week 4	19/3	Iterations
Week 5	26/3	Functions & Recursion
Week 6	2/4	Lists, Tuples, Sets, Dictionaries
Week 7	9/4	Numpy Arrays
Week 8	16/4	Strings
Week 9	23/4	Comprehensive Review
	Week 10	30/4 Midterm Exam
	Week 11	7/5 Pandas DataFrames
	Week 12	14/5 Matplotlib Visualizations
	Week 13	21/5 Processing Time Series Data
	Week 14	28/5 Processing Image Data
	Week 15	4/6 Comprehensive Review
	Week 16	11/6 Final Exam

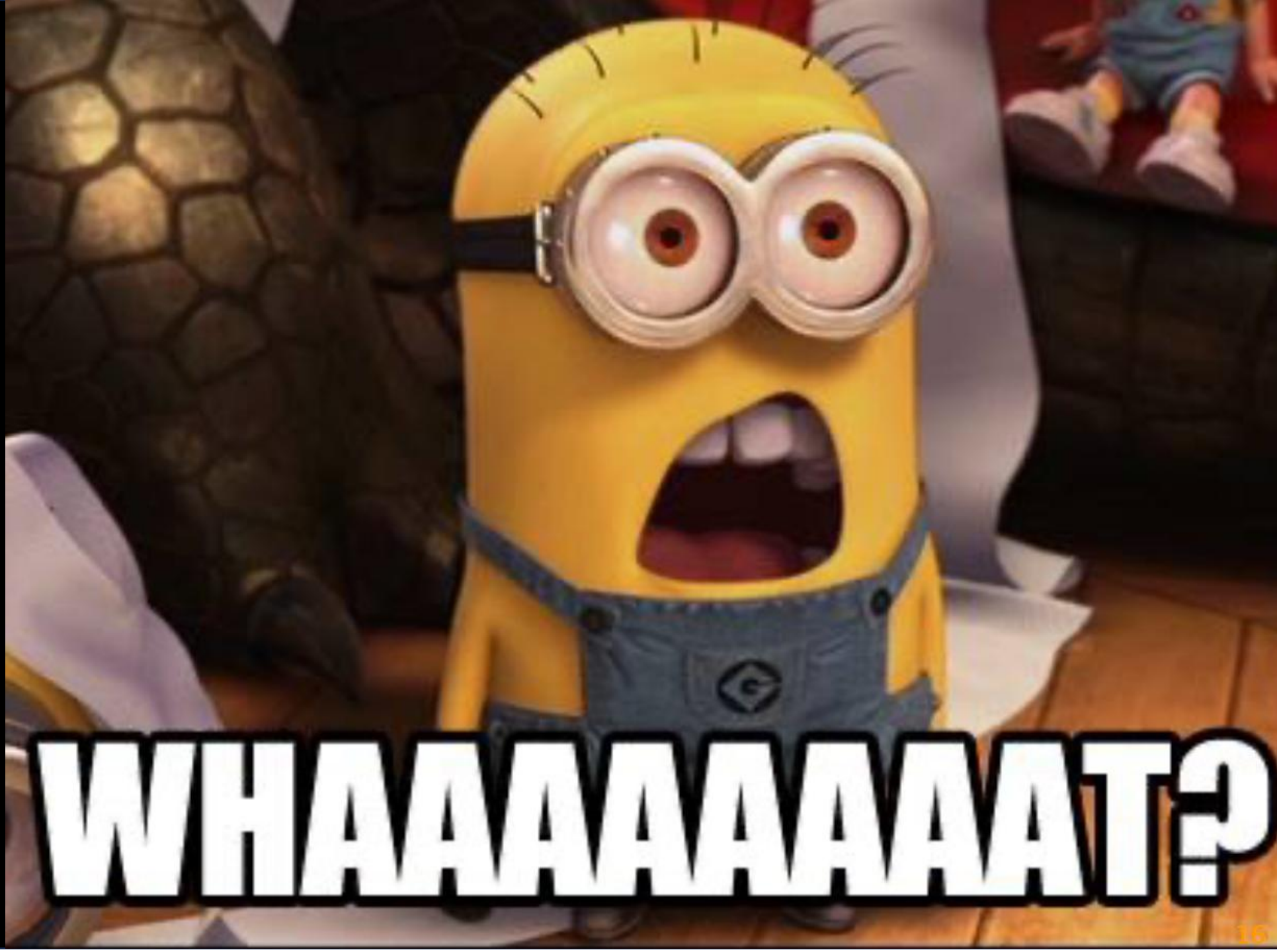
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Who teaches you programming

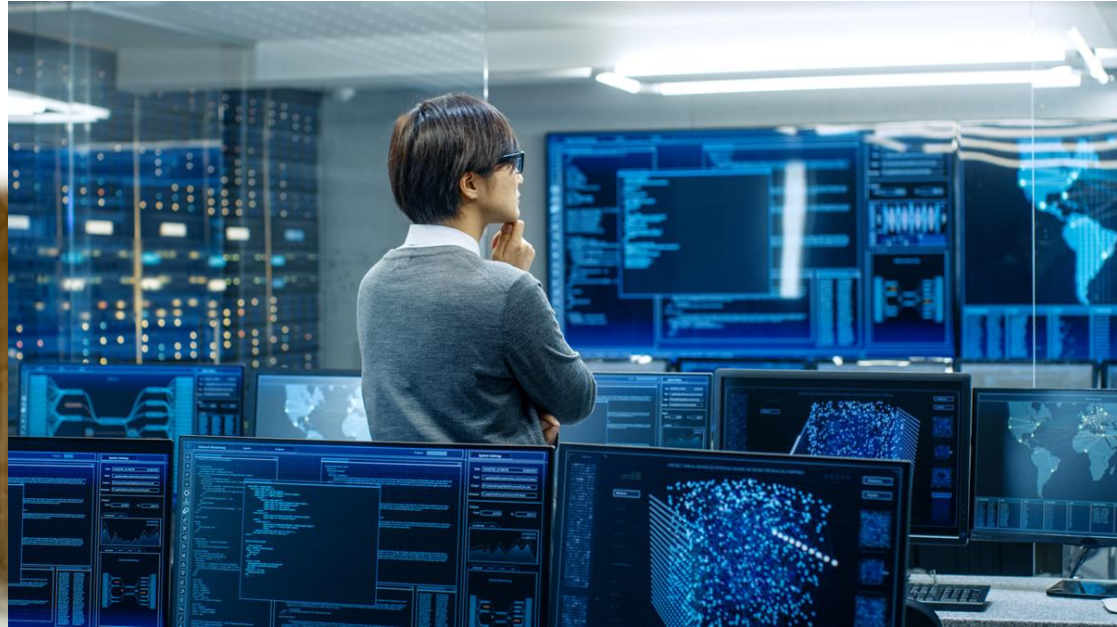
- An engineer

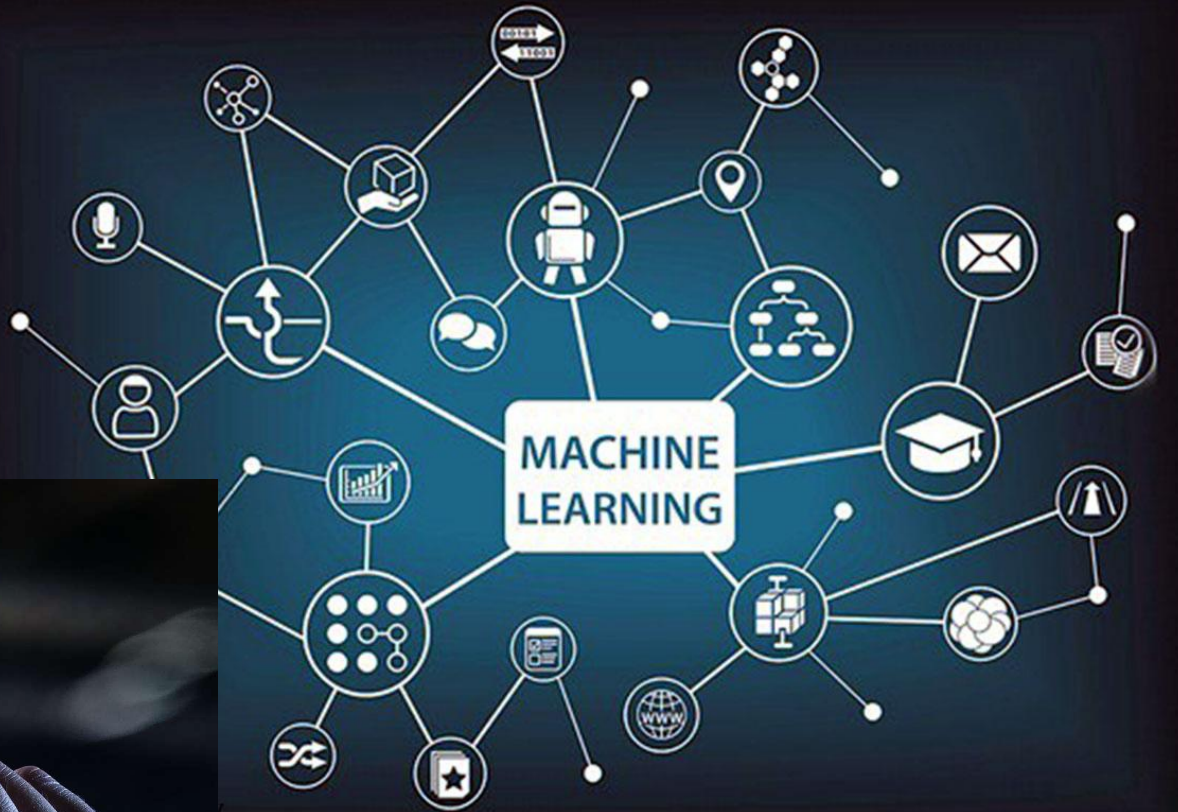
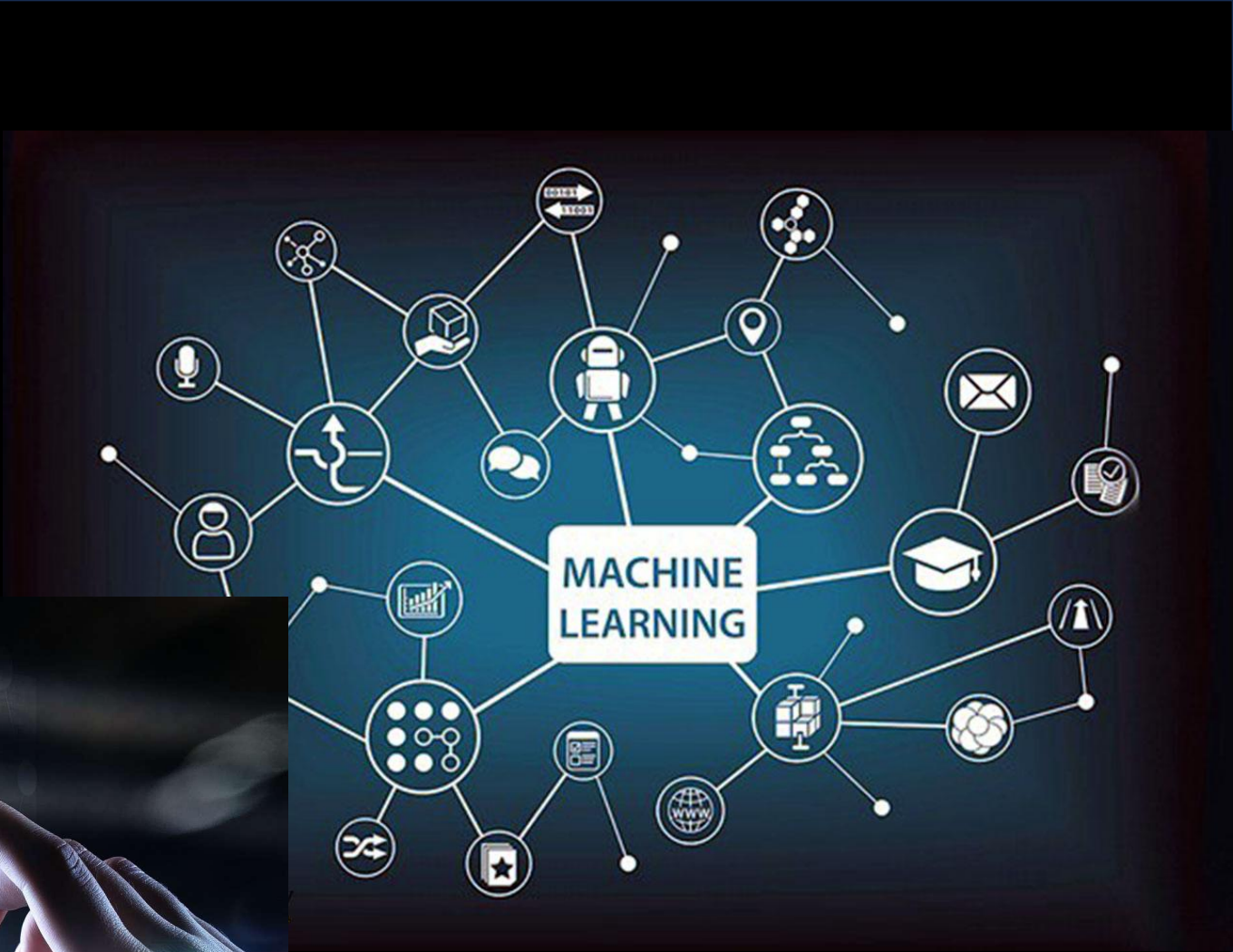


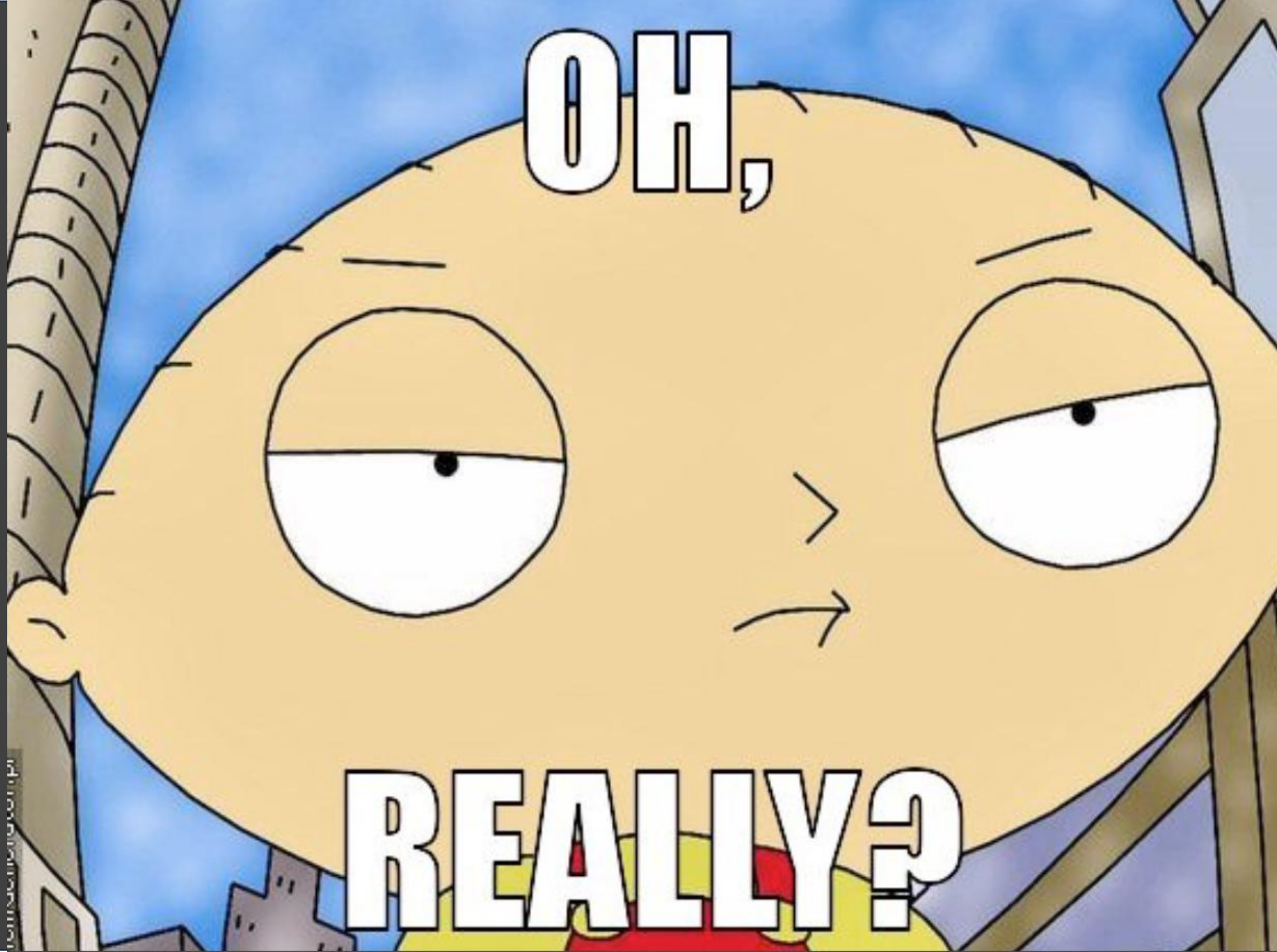


Who teaches you programming

- An engineer
- Much better







GeoAI

Interdisciplinary domain at the confluence of **artificial intelligence (AI)**, **data science (DS)**, and **geographic information systems (GIS)**

Applied expansion of AI, teaching machines to gain the intelligence of automatically performing **spatial analysis tasks**

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Examination

The grading (100 points) is split into:

- 1st part – elements of programming (50/100 points)
- 2nd part – programming for data analysis (35/100 points)

Examination

The grading (100 points) is split into:

- **1st part – elements of programming (50/100 points)**
 - 2nd part – applied programming (35/100 points)
-
- 50 points are assigned through a mid-term examination (Week 10, April 30th)

Examination

The grading (100 points) is split into:

- 1st part – elements of programming (50/100 points)
 - **2nd part – applied programming (35/100 points)**
-
- 35 points are assigned through a final examination (Week 16, June 11th)

Examination

The grading (100 points) is split into:

- 1st part – elements of programming (50/100 points)
- 2nd part – applied programming (35/100 points)
- **15 additional points are assigned through homework**

TA and Office Hours

Teaching Assistant:

- Chi-Chun Chiou (“Ken”) (b10103022@ntu.edu.tw)

Office Hours (Ken):

(arranged by email first)

Books that may be used as general references are the following:

- Severance (2016), Python for Everybody: Exploring Data in Python 3, CreateSpace Independent Publishing Platform.
- Deitel and Deitel (2021), Intro to Python for Computer Science and Data Science, Pearson FT Press
- McKinney (2017), Python for Data Analysis, 2nd edition, O'Reilly Media.

Anyone Who Has Never
Made a

Mistake

Has Never Tried
Anything
New.

Albert Einstein

A scene from the movie 'The Avengers' showing Tony Stark (Robert Downey Jr.) and Bruce Banner (Mark Ruffalo) in a sleek, futuristic car. Tony is in the driver's seat, looking forward, while Bruce is in the passenger seat, looking out the window. The car is dark and highly reflective, with a glowing red light strip visible on the side. The background is a blurred city street at night.

GET IN LOSERS

**WE'RE GOING TO
DO SCIENCE.**

See you
next week!